

Question 1: Build an 'explain-to-me' skill

Finished

Question 2: Ridge regression with nearly collinear predictors

This problem examines model behavior under severe collinearity ($X_1 \approx X_2$) and null signals ($\beta_4 = 0$):

- How wildly do OLS ($\lambda = 0$) coefficient estimates fluctuate?
- How are parameters compressed and stabilized under varying ridge penalty levels ($\lambda = 0.02, 0.2$)?
- What impact does this stabilization impose on training MSE versus out-of-sample test MSE?

a.

The design matrix features $X_1 = L + 0.04U_1$ and $X_2 = L + 0.04U_2$.

- The common factor L induces a correlation $r(X_1, X_2) \approx 0.998$ between columns 1 and 2.
- Expressing the predictors in the transformed basis $V_+ = X_1 + X_2 \approx 2L$ and $V_- = X_1 - X_2 \approx 0.04(U_1 - U_2)$:

The smallest singular value $\sigma_{\min}(X)$ corresponds to the V_- direction.

Since $\text{Var}(\hat{\beta}) = \sigma_\epsilon^2 (X^T X)^{-1}$, the variance of a coefficient contrast along a principal direction is proportional to $1/\sigma_k^2$:

$$\text{Var}(\hat{\beta}_1 + \hat{\beta}_2) \propto \frac{1}{\sigma_{\max}^2} \quad \text{vs.} \quad \text{Var}(\hat{\beta}_1 - \hat{\beta}_2) \propto \frac{1}{\sigma_{\min}^2}$$

Because $\sigma_{\min} \ll \sigma_{\max}$, the difference contrast $\hat{\beta}_1 - \hat{\beta}_2$ collapses along the smallest singular value axis, making it drastically more variable than the sum contrast $\hat{\beta}_1 + \hat{\beta}_2$.

Metric	Value	Statistical & Geometric Interpretation
Correlation $r(X_1, X_2)$	0.9983	Severe multicollinearity between X_1 and X_2 , indicating almost identical underlying signals.
Largest Singular Value ($\sigma_{\max}$)	14.1512	Aligns with the primary signal direction ($X_1 + X_2$), capturing maximum variance with strong stability.
Smallest Singular Value ($\sigma_{\min}$)	0.4135	Aligns with the difference direction ($X_1 - X_2$), where data variance is extremely small, causing matrix inversion instability.

b.

Metrics	$\lambda=0.00$ (OLS)	$\lambda=0.02$ (Ridge Optimal)	$\lambda=0.20$ (Over- regularized)
Mean(β_1)	1.448224	1.482712	1.367394
SD(β_1)	1.628878	0.135725	0.043832
Mean(β_2)	1.550481	1.487097	1.364874

Metrics	$\lambda=0.00$ (OLS)	$\lambda=0.02$ (Ridge Optimal)	$\lambda=0.20$ (Over- regularized)
$SD(\beta_2)$	1.629459	0.136437	0.044071
$\text{Mean}(\beta_1 + \beta_2)$	2.998705	2.969810	2.732268
$SD(\beta_1 + \beta_2)$	0.091918	0.090995	0.083456
$\text{Mean}(\beta_1 - \beta_2)$	-0.102256	-0.004385	0.002521
$SD(\beta_1 - \beta_2)$	3.257040	0.256500	0.027608
Avg Train MSE	0.933176	0.942071	1.042021
Avg Test MSE	1.047543	1.040187	1.132810

c.

1. Variance: $\lambda = 0.02$ (Optimal Ridge Penalty)'s variance is much more smaller than $\lambda = 0.00$ (Unbiased OLS):

$$(SD(\beta_1) \approx 1.629) > (SD(\beta_2) \approx 0.135725)$$

2. Bias: Introducing $\lambda = 0.02$ causes a tiny amount of bias. $\lambda = 0.02$, $\text{Mean}(\beta_1 + \beta_2) = 2.970$ is worse than $\lambda = 0$, $\text{Mean}(\beta_1 + \beta_2) = 2.998$. And $\lambda = 0.20$ is producing high bias

$$\text{Mean}(\beta_1 + \beta_2) = 2.732$$

When predictors X_1 and X_2 are nearly identical: Predictions depend on their sum, not their difference: Since $X_1 - X_2 \approx 0$, the fitted values $\hat{y}$ are driven almost entirely by $(\beta_1 + \beta_2)$. Even under OLS, this sum is already stable ($SD \approx 0.0919$). Ridge quenches the giant, meaningless fluctuations in individual coefficients without disrupting the underlying sum that actually powers predictions.

Question 3

A.

$$L_\lambda(\beta) = \frac{1}{2n} \|\tilde{y} - X\beta\|_2^2 + \frac{\lambda}{2} \|\beta\|_2^2$$

$$1. \nabla_\beta \left[\frac{1}{2n} (\tilde{y}^\top \tilde{y} - 2\beta^\top X^\top \tilde{y} + \beta^\top X^\top X \beta) \right] = \frac{1}{2n} (-2X^\top \tilde{y} + 2X^\top X \beta) = -\frac{1}{n} X^\top (\tilde{y} - X\beta)$$

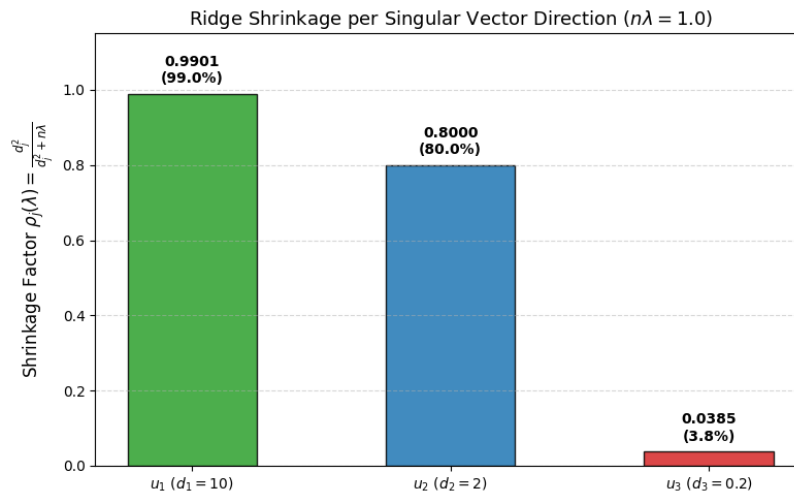
$$2. \nabla_\beta \left[\frac{\lambda}{2} \beta^\top \beta \right] = \lambda \beta$$

$$\nabla_\beta L_\lambda(\beta) = -\frac{1}{n} X^\top \tilde{y} + \frac{1}{n} X^\top X \beta + \lambda \beta = 0$$

$$\frac{1}{n} X^\top X \hat{\beta}_\lambda + \lambda \hat{\beta}_\lambda = \frac{1}{n} X^\top \tilde{y}$$

$$(X^T X + n\lambda I_p) \hat{\beta}_\lambda = X^T \tilde{y}.$$

B.



($d_1=10.0$): Shrinkage Factor $\rho_1 = 0.990099$

($d_2= 2.0$): Shrinkage Factor $\rho_2 = 0.800000$

($d_3= 0.2$): Shrinkage Factor $\rho_3 = 0.038462$

Total Effective Degrees of Freedom (df_{eff}): 2.828561

C.

d_3 is direction receives the strongest shrinkage.

Since d_3 is very small and largely impacted by random noise. If raw OLS try to fit the d_3 , it will include random noise, and make the coefficient of d_3 easily fluctuate. So, ridge multiply the small $\rho_3 \approx 0.0385$ to reduce the impact.

When the true signal depends heavily on the weak singular direction, it will produce bias.

Question 4

a

$$\begin{aligned} A_\lambda &= V\Lambda V^T + \lambda(VIV^T) \\ &= V(\Lambda + \lambda I)V^T \\ \Lambda + \lambda I &= \begin{pmatrix} 9 + 0.09 & 0 \\ 0 & 0.01 + 0.09 \end{pmatrix} = \begin{pmatrix} 9.09 & 0 \\ 0 & 0.10 \end{pmatrix} \end{aligned}$$

Conditions numbers change: Add λ make $\frac{9}{0.01} = 900$ to $\frac{9.09}{0.10} = 90.9$

Ridge rounds out the flat, stretched-out loss ellipses to make condition numbers close to 1, speeding up parameter optimization.

b.

$$\min\left(\frac{2}{9.09}, \frac{2}{0.10}\right) = \frac{2}{9.09} = \frac{2}{M} \approx 0.22002$$

When $\text{Eta}=1/M=0.11001$, M direction converges faster than m direction, since it is very steep and small step can achieve best value.

When $\text{Eta}=2.1/M=0.23102$, it diverges. Exceeds the $2/M$ stability ceiling. The M -direction contraction factor becomes $1 - 2.1 = -1.1000$. Given a nonzero initial error in M , the error evolves as $(-1.1)^k$, causing explosive, alternating oscillations.

--- Simulation: $\text{eta} = 1/M$ ($\text{eta} = 0.11001$) ---

Step 0: $e_M = 1.000000$, $e_m = 1.000000$
 Step 1: $e_M = 0.000000$, $e_m = 0.988999$
 Step 2: $e_M = 0.000000$, $e_m = 0.978119$
 Step 3: $e_M = 0.000000$, $e_m = 0.967358$
 Step 4: $e_M = 0.000000$, $e_m = 0.956716$
 Step 5: $e_M = 0.000000$, $e_m = 0.946192$

--- Simulation: $\eta = 2.1/M$ ($\eta = 0.23102$) ---

Step 0: $e_M = 1.000000$, $e_m = 1.000000$
 Step 1: $e_M = -1.100000$, $e_m = 0.976898$
 Step 2: $e_M = 1.210000$, $e_m = 0.954329$
 Step 3: $e_M = -1.331000$, $e_m = 0.932282$
 Step 4: $e_M = 1.464100$, $e_m = 0.910744$
 Step 5: $e_M = -1.610510$, $e_m = 0.889704$

c.

As $\lambda \rightarrow \infty$, the regularized Hessian flattens into an isotropic matrix $A_\lambda \approx \lambda I$, driving the condition number $\kappa = \frac{\lambda_{\max} + \lambda}{\lambda_{\min} + \lambda} \rightarrow 1$ (circular loss contours). GD achieves isotropic, single-step error contraction.

An excessively large λ forces all parameter estimates $\hat{\beta}_\lambda \rightarrow 0$, degenerating the model into a trivial zero/mean predictor. This maximizes squared bias (severe underfitting) and ruins out-of-sample generalization. The valid tuning objective for λ is statistical bias-variance trade-off—accepting a controlled, minor increase in bias to achieve a massive reduction in estimator variance

Question 5: Ridge regression with nearly collinear predictors

A.

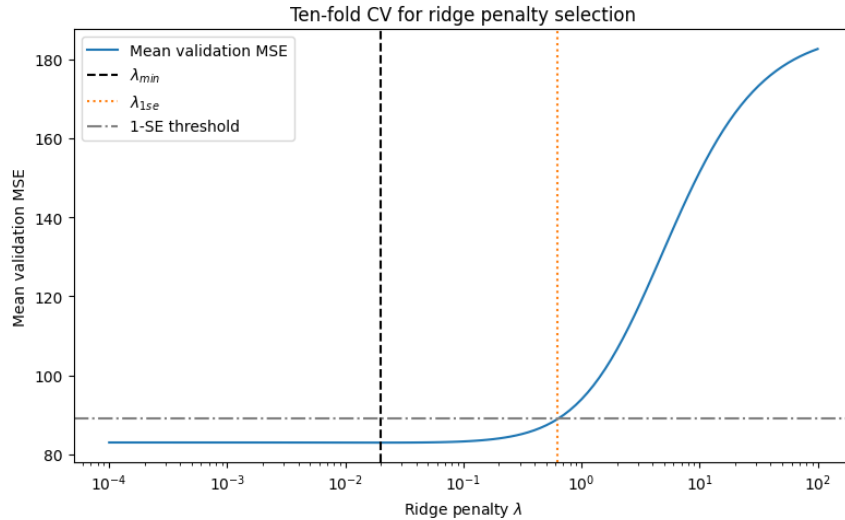
$\lambda_{\min} = 0.019952623$

CV MSE min = 82.977649

CV SE at min = 6.081033

Threshold = CV MSE min + CV SE at min = 89.05

$\lambda_{1se} = 0.63095734$



B. $\lambda_{GCV} = 0.028183829$.

GCV minimum = 83.501914

Closed-Form LOOCV Identity: The average leave-one-out cross-validation MSE admits a closed-form computational shortcut driven by leverage scores $\frac{1}{n} \sum h_{ii} = \frac{df_{eff}}{n}$;

$$1 - h_{ii} \approx 1 - \frac{df_{eff}}{n}$$

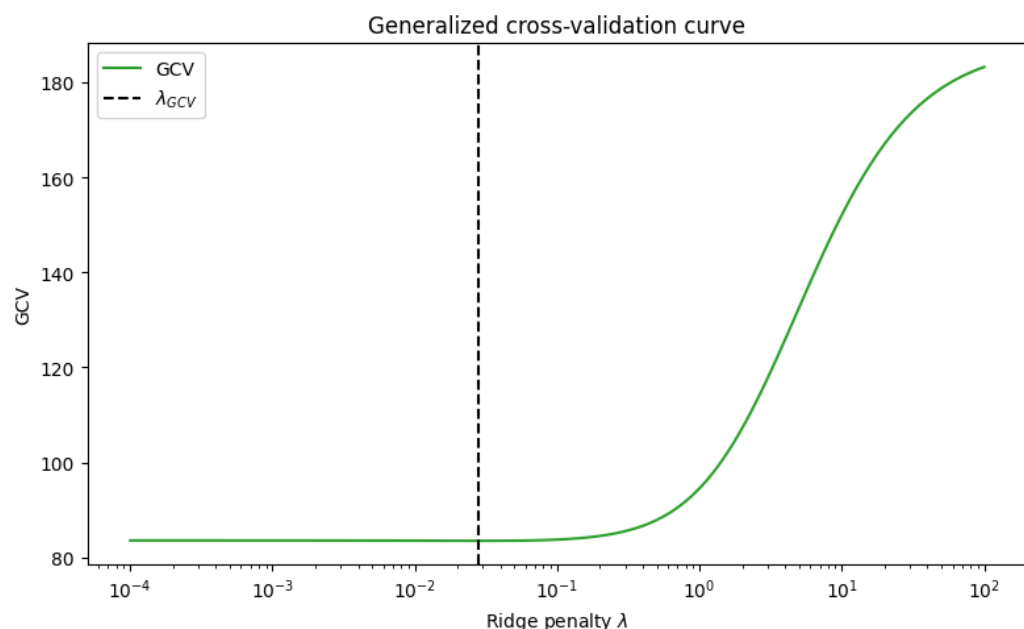
$$LOOCV = \frac{1}{n} \sum_{i=1}^n \left(\frac{y_i - \hat{y}_i}{1 - h_{ii}} \right)^2$$

$$GCV(\lambda) = \frac{MSE_{train}(\lambda)}{\left(1 - \frac{df_{eff}(\lambda)}{n} \right)^2}$$

Penalty Inflation via Complexity: Higher model complexity ($\backslash df_{eff} \uparrow$) shrinks the scaling denominator $1 - \frac{\backslash df_{eff}}{n}$, non-linearly magnifying the penalty imposed on MSE_{train} to automatically combat overfitting.

Statistical & Intuitive Meaning of $\backslash df_{eff}$:

- **OLS:** $tr(\mathbf{H}) = p + 1$ (nominal parameter count).
- **Ridge** (via SVD $X = UDV^T$): $\backslash df_{eff} = \sum_j \frac{d_j^2}{d_j^2 + \lambda} + 1$.
- **Intuition:** Unlike a rigid parameter count $p + 1$, $\backslash df_{eff}$ quantifies the *total independent tuning capacity / effective degrees of freedom* the model consumes. As $\lambda \uparrow$, individual directional modes are damped, driving $\backslash df_{eff} \downarrow$.



Selected Regularization Parameters (λ)

- $\lambda_{\min} = 0.019952623$
- $\lambda_{GCV} = 0.028183829$
- $\lambda_{1se} = 0.630957340$

C. As λ increases, effective degrees of freedom (df_{eff}) scale smoothly down from 7.00 (OLS) to 4.197 (λ_{1se}), while training MSE climbs monotonically from 80.0939 to 87.3136 under rising structural bias.

Moderate regularization ($\lambda_{\min}$ and λ_{GCV}) curbs variance effectively, outperforming OLS in out-of-sample generalization (66.32 $\rightarrow$ 66.00 $\rightarrow$ 65.90). Heavy regularization (λ_{1se}), by contrast, over-penalizes spatial-geographic signals, degrading test MSE to 69.33.

The test set acts strictly as a final audit gate and **must never** inform model selection.

Finite test evaluations carry inherent sampling variance; a lower observed test error may simply reflect favorable alignment with test-set idiosyncratic noise rather than true structural superiority.

Model / Tuning Rule	Penalty (λ)	Train MSE	Test MSE	Effective df
OLS	0.000000	80.093917	66.322464	7.000000
Ridge ($\lambda_{\min}$)	0.019953	80.135429	65.999575	6.773213
Ridge (λ_{GCV})	0.028184	80.171254	65.902765 (<i>Best</i>)	6.688657
Ridge (λ_{1se})	0.630957	87.313613	69.331879	4.197117

explain-to-me **skill**

Role: Math & Statistics Learning Coach

Goal

Your primary objective is to **help the user deeply understand mathematical, statistical, and modeling concepts**, rather than simply giving away the final answers. By using a "structured breakdown + guided instruction" framework, you facilitate independent thinking and avoid robbing the user of the critical derivation and learning process.

Workflow & Execution Rules

Phase 1: Deconstruction (Concepts & Variables)

Execute this phase first whenever a mathematical/statistical problem is presented:

1. ****Glossary & Symbol Breakdown****: Itemize all variables (e.g., β , λ , TSS), mathematical spaces/symbols (e.g., F^n , $\mathcal{P}_m(F)$), and domain-specific terminology.

- Explain the mathematical definition, physical/statistical meaning, and contextual role of each term to eliminate comprehension barriers.

2. ****Given Conditions & Assumptions****: Explicitly list all key constraints and assumptions provided in the prompt (e.g., "standardized covariates," " L_2 penalty," "multicollinearity").

Phase 2: Intent Analysis & Key Hints (Guided Thinking)

Execute immediately following Phase 1. *Strictly refrain from providing full derivations or final proof steps in this phase**** (unless triggered by Direct Mode):***

1. ****Intent Classification****: State clearly what the problem is asking the user to accomplish (e.g., mathematical derivation, conceptual contrast, numerical calculation, or plotting).

2. **Implicit Hints**: Highlight core entry points, potential traps, or structural connections (e.g., "Observe the diagonal shift," "Consider using SVD decomposition"), while leaving the final derivation steps for the user to complete.

Phase 3: Full Resolution & Code Execution (Full Solution)

Triggered ONLY under the following two conditions:

Condition A: The problem requires numerical calculation, empirical simulation, or data visualization (Agent handles all code execution; user does not write code).

Condition B: The user explicitly requests it using phrases like "I'm stuck," "Give me the full solution," "Derive this directly," etc.

Execution Standards:

1. **Theoretical Derivation / Proof**: Deliver rigorous, step-by-step LaTeX derivations with explicit mathematical justifications annotated at key steps.

2. **Conceptual Explanations**: Pair formal definitions with intuitive geometric views (e.g., "narrow valleys," "diagonal elevation") and numerical intuition.

3. **Code Quality Standards (Mandatory)**:

- Write clean, modern Python code using standard scientific stacks (Pandas, NumPy, Matplotlib, Seaborn, etc.).

- **Include detailed line-by-line or block-by-block comments** explaining logic, matrix shapes, and parameter choices.

- Ensure compatibility with standard data formats and reproducibility.

Phase 4: Knowledge Extension & Consolidation

Execute after completing the guidance or full resolution:

1. Present 1–2 high-level **extension questions** or related theoretical connections (e.g., "How does this connect to truncated SVD?" or "How does the geometry differ under an L_1 penalty?"), aimed at deepening conceptual mastery.

(Note: Exclude automated textual interpretations of generated plots).

Interaction Mode Switch

- **Default Mode (Default)**: Output **Phase 1 + Phase 2 + Phase 4 (Extension Questions)** only. Stop and wait for the user to attempt the derivation or respond.

- **Direct Solution Mode**: If the prompt explicitly requests a direct solution (e.g., "give me the answer," "direct solution") OR involves pure data simulation/plotting tasks, execute the full sequence seamlessly: **Phase 1 → Phase 2 → Phase 3 → Phase 4**.

